1. Introduction

Large language models have reshaped natural language processing by learning representations directly from raw text. This paper studies a family of architectures built on self-attention, which replace recurrence with direct pairwise interactions between tokens. We argue that attention alone is sufficient for strong performance on sequence tasks, and we present empirical results across translation and language modeling benchmarks.

2. The Attention Mechanism

Attention computes a weighted sum of value vectors, where each weight reflects the similarity between a query and its corresponding key. Given queries Q, keys K, and values V, the scaled dot-product attention is defined as softmax(Q K^T / sqrt(d_k)) V, where d_k is the key dimension. Dividing by the square root of the dimension keeps gradients stable as the dimensionality grows. Multi-head attention runs several attention operations in parallel and concatenates their outputs, allowing the model to attend to information from different representation subspaces.

3. Transformers

The Transformer stacks attention layers together with position-wise feedforward networks. Each sub-layer is wrapped with a residual connection followed by layer normalization. Because attention is permutation invariant, positional encodings are added to the input embeddings so the model can reason about order. Encoder layers attend over the input sequence, while decoder layers use masked attention to prevent information from flowing from future positions.

4. Scaling Laws

Scaling laws describe how model loss decreases as a power law in compute, dataset size, and parameter count. Empirically, loss falls predictably as each of these factors grows, which allows practitioners to forecast the benefit of larger training runs. These findings motivate training large models on vast corpora and then adapting them to downstream tasks through fine-tuning or prompting.

5. Conclusion

We presented an attention-based architecture that dispenses with recurrence and convolution. The model trains faster, parallelizes well over sequence positions, and reaches strong results on translation and language modeling. Attention provides a general mechanism for modeling dependencies, and we expect it to remain a core building block for future sequence models.